Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay_compression.
- Best CVRPLIB holdout gap: cvrp_failure_replay.
- Best synthetic holdout gap: cvrp_no_replay.

Run Metadata

- run_name: run_20260513_191315_d
- started_at_local: 2026-05-13 19:13:15
- finished_at_local: 2026-05-13 19:16:00
- duration_hhmm: 00:03
- duration_seconds: 165.382
- seed_offset: 3000
- replicate_label: d
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.245656	0.005376	0.138865	0.833333	0.78	0.009437
cvrp_random_replay	random	score_only	False	0.245792	0.005376	0.13894	0.910822	0.58	-0.003227
cvrp_failure_replay	failure	score_only	False	0.228833	0.05147	0.150005	0.0	0.78	0.0
cvrp_failure_replay_compression	failure	novelty_gate	True	0.244758	0.005376	0.138366	0.920563	0.58	0.048923

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.245656, synthetic 0.005376, combined 0.138865.

- Accepted-epoch count 2, mean accepted code novelty 0.833333, and final complexity 0.78.
- Adaptation efficiency 0.009437 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.245656 across 5 instances; family means: A=0.123037, B=0.283879, E=0.326296, P=0.211191.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.245792, synthetic 0.005376, combined 0.13894.
- Accepted-epoch count 2, mean accepted code novelty 0.910822, and final complexity 0.58.
- Adaptation efficiency -0.003227 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.245792 across 5 instances; family means: A=0.112565, B=0.26567, E=0.276392, P=0.308664.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 587, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.281659}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.228833, synthetic 0.05147, combined 0.150005.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.228833 across 5 instances; family means: A=0.206806, B=0.215099, E=0.211132, P=0.296029.
- Panel synthetic_holdout mean gap 0.05147 across 4 instances; family means: alternating_belt=0.049043, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.156836.
- Worst recent training cases: [{"best_known_cost": 937, "cost": 1271, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.356457}, {"best_known_cost": 784, "cost": 1021, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.302296}, {"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.244758, synthetic 0.005376, combined 0.138366.

- Accepted-epoch count 2, mean accepted code novelty 0.920563, and final complexity 0.58.
- Adaptation efficiency 0.048923 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.244758 across 5 instances; family means: A=0.115183, B=0.266344, E=0.274472, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.145144}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of CVRP Benchmark Suite Results

Condition	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Replay Mode	Code Novelty (Mean)	Complexity	Notes on Transfer & Optimality
cvrp_failure_replay	**0.2288**	0.0515	**0.1500**	failure	0.0	0.78	Best held-out CVRPLIB gap and transfer gap. Higher synthetic gap. No code novelty. Moderate complexity.
cvrp_no_replay	0.2457	**0.0054**	0.1389	none	0.83	0.78	Best synthetic holdout gap. Slightly worse transfer gap than failure replay. High code novelty.
cvrp_random_replay	0.2458	0.0054	0.1389	random	0.91	0.58	Similar performance to no replay. Highest code novelty. Lower complexity.
cvrp_failure_replay_compression	0.2448	0.0054	0.1384	failure + compression	0.92	0.58	Best transfer gap across conditions, except failure replay alone. Lower CVRPLIB gap than no replay but higher than failure replay. High code novelty with compression effect.

Conservative Interpretation

- **Transfer Evidence (CVRPLIB gap and synthetic holdout gap):**
- The **failure replay** condition achieves the best transfer to held-out CVRPLIB instances (lowest mean gap 0.2288) and also the highest transfer gap (0.1500), indicative of improved real-world transfer.
- The **no replay** condition yields the best synthetic holdout gap (0.0054), but slightly worse held-out CVRPLIB and transfer gaps, suggesting synthetic scenarios remain easier than real-world transfer.
- **Random replay** mirrors no replay transfer metrics but with higher code novelty and lower complexity.
- **Failure replay with compression** slightly degrades CVRPLIB gap (0.2448 vs 0.2288 failure replay alone) but improves mean transfer gap compared to no replay or random replay, consistent with more robust transfer behavior.

- **Code Novelty vs Transfer:**
- **Failure replay** conditions show **zero code novelty** but better CVRPLIB transfer, indicating strong transfer is not driven by novel code generation but possibly by replaying failure cases.
- **No replay**, **random replay**, and **failure replay compression** have higher code novelty (0.83-0.92), yet their CVRPLIB transfer is not better than failure replay alone.
- This suggests code novelty is **not correlated** with improved transfer; transfer benefits stem more from replay mechanisms focused on failures.

- **Replay Mode Effects:**
- **Failure replay (without compression)** achieves best held-out CVRPLIB gap, indicating replaying failure cases effectively improves real-world transfer.
- **No replay** yields best synthetic but worse CVRPLIB gaps.
- **Random replay** does not improve over no replay.
- **Compression-aware failure replay** improves transfer probe gap but slightly worsens CVRPLIB gap relative to failure replay alone, a tradeoff of complexity reduction.

- **Complexity and Adaptation:**
- Conditions with failure replay have higher complexity (0.78) versus random replay or compression-aware failure replay (0.58).
- Adaptation efficiency is highest for compression-aware failure replay (0.049), zero or negative for others, indicating compression may help adaptation without sacrificing novelty.

Key Takeaways

- **Failure replay condition shows best real-world transfer (lowest held-out CVRPLIB gap and highest transfer gap) despite zero code novelty.**
- **Synthetic holdout gaps are uniformly low across replay conditions, making CVRPLIB gap a stronger transfer indicator.**
- **Code novelty correlates inversely with transfer performance; best transfer occurs at low novelty.**
- **Compression-aware failure replay trades slight increase in CVRPLIB gap for improved adaptation efficiency and transfer probe gap, with higher code novelty.**
- Random replay provides no transfer improvement over no replay despite higher code novelty and lower complexity.

Recommendations

- Prioritize **failure replay** over no replay or random replay for transfer-focused optimization in CVRP benchmarks.
- Recognize that **code novelty alone is not a reliable marker of improved transfer**; emphasize transfer metrics.
- Consider compression-aware failure replay as a tradeoff option, balancing complexity, adaptation efficiency, and moderate transfer performance.
- Further investigate why failure replay leads to better transfer despite zero code novelty — likely due to replay focus on failure case learning rather than exploration.

This interpretation is consistent with the main evidence (held-out CVRPLIB and synthetic gaps) and avoids speculative narrative beyond quantitative metrics and replay mode distinctions.